

Sabharish P V

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Professional Summary

Computer Science undergraduate with experience in Software Development, Machine Learning, and LLM-based applications. Built research-oriented and production-focused systems in context retrieval, management, and conversational AI. Interested in AI and intelligent software systems.

Experience

Backend Developer Intern

Jan 2026 – Apr 2026

Geojit Technologies

- Built a Drools-based decision engine (Java 21, Spring Boot) for trading account validation, using Drools' AGENDA-GROUP architecture to isolate rule execution paths and reduce processing overhead
- Designed a custom diagnosis service that traces failed validations against parsed Excel rule metadata to return specific, human-readable failure reasons
- Built a React admin frontend for live in-browser editing of Excel-based decision tables (via Apache POI), with automated pre-edit snapshotting and one-click rollback, and a KPI dashboard for engine/rule status

Research Intern – IEEE CIS Kerala Section Summer Internship 2025

May 2025 – Jul 2025

Cochin University of Science and Technology, hosted by IEEE Computational Intelligence Society

- Built a graph-based facial expression recognition pipeline: extracted 468 facial landmarks per frame via MediaPipe Face Mesh, encoded them as weighted graphs, and modeled temporal edge-weight variation with an LSTM for video-based emotion classification
- Achieved 93.25% test accuracy (95.22% macro precision, 94.97% macro recall) across 8 emotion classes on the RAVDESS dataset, with class-weighted loss to handle label imbalance
- Cut redundant computation across training runs by building a caching layer into the preprocessing pipeline

Projects

MakeSense – Optimal Transport Context Retrieval Engine

Winner, YIP 8.0 – Research Track

Training-free Python library that reformulates retrieval as set-level semantic coverage using Optimal Transport

- Designed an Optimal Transport-based context construction algorithm that selects the precise combination of facts needed to answer complex, multi-hop questions, rather than nearest-neighbor similarity retrieval
- Evaluated across HotpotQA, FEVER, and SciFact, improving Hit@5 by $9.5\times$ over dense retrieval baselines on HotpotQA while reducing adaptive context size by over 90%
- Advanced to the final round of the IEEE InApp Student Project Awards 2026 (Project IIA-2083)
- Co-authored a paper on this work, currently under peer-review at *Knowledge and Information Systems* (Springer)

Eviot – Conversational Reasoning Application

FastAPI & Next.js application built on the MakeSense engine for per-turn context-set construction in conversation

- Implemented SSE-streamed responses across query resolution, OT-based evidence selection, and answer generation, surfacing each stage of retrieval live
- Built a persistent memory system (fact extraction, reconciliation, structured storage) that resolves pronouns and references across turns for multi-turn continuity, visualized as an interactive force-directed knowledge graph with a toggleable editor for manual correction

Education

B.Tech. in Computer Science and Engineering

Nov 2022 – May 2026

Rajagiri School of Engineering and Technology

CGPA: 9.18 / 10

Technical Skills

Languages: Python, Java, C, C++, SQL

Frameworks & Tools: Spring Boot, Drools, FastAPI, LangChain, LangGraph, Git

Certifications: NPTEL Machine Learning, NPTEL Deep Learning, Machine Learning Specialization (DeepLearning.AI)